

CSA0311 DATA STRUCTURESASSESSMENT TOOL -1ANALYTICAL PROBLEM

1. Convert the following infix expression into postfix notation using a stack. Show the contents of the stack after processing each symbol.

$$(A+B) \times (C-D) / E$$

Infix expression,

Ans
2017

CHARACTER	STACK	OUTPUT
(	(	Empty
A	(	A
+	(+	A
B	(+	AB
)	(+)	AB +
*	*	AB +
(	* (	AB +
C	* (	AB + C
-	* (-	AB + C
D	* (-	AB + CD
)	* (-)	AB + CD -
/	/	AB + CD - /
E	/	AB + CD - * E

Ex 8

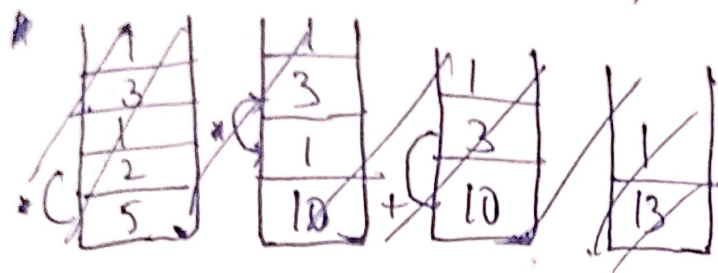
$$AB + CD - E /$$

i) Final Postfix Expression.

$$AB + CD - E /$$

ii) Postfix evaluation

$$A = 5, B = 2, C = 1, D = 3, E = 1$$



$$i) 5 + 2 = 7$$

$$ii) 1 - 3 = -2$$

$$iii) 7(-2) = -14$$

$$\downarrow$$

$$-14 \div 1 = -14$$

2. Convert the following infix expression into postfix form. Explain how operator precedence and associativity influence the conversion.

$$A + B * C - (D / E + F) * G$$

Infix expression,

$$A + B * C - (D / E + F) * G$$

CHARACTERS	STACK	POSTFIX
A		A
+	+	A
B	+	AB
*	+*	AB
C	+*	ABC
-	-	ABC*

(	- (	$ABC^* +$
D	- (	$ABC^* + D$
/	- (/	$ABC^* + D$
E	- (/	$ABC^* + DE$
+	- (+	$ABC^* + DE /$
F	- (+	$ABC^* + DE / F +$
)	- (+)	
*	- *	$ABC^* + DE / F +$
*	- *	$ABC^* + DE / F + G$
G		$ABC^* + DE / F + G^* -$
Empty		

Final Postfix Expression,

$ABC^* + DE / F + G^* -$

Operator Precedence,

- i) Parentheses () $\rightarrow$ Highest priority
- ii) Multiplication $*$ and Division $/$
- iii) Addition $+$ and Subtraction $-$

- $+$, $-$, $*$, $/$ are left to right associative
- When two operators have the same precedence, the left operator is evaluated first.